L8 22/09/2020

(A,B) controllable $A \in \mathbb{R}^{n\times n}$ $B \in \mathbb{R}^{n\times m}$ $B = [b_1\ b_2 \cdots b_m]$

Step 1:

$$L := [\underbrace{b_1\ Ab_1\ A^2b_1 \cdots A^{d_1-1}b_1}_{L_1 \in \mathbb{R}^{n\times d_1}}\ \underbrace{b_2\ Ab_2 \cdots A^{d_2-1}b_2}_{L_2 \in \mathbb{R}^{n\times d_2}} \cdots \underbrace{b_m \cdots A^{d_m-1}b_m}_{L_m \in \mathbb{R}^{n\times d_m}}] \in \mathbb{R}^{n\times n}$$

$\{d_1, d_2 \ldots d_m\} \longrightarrow$ controllability indices

$\mu := \max_{1\le i\le m}\{d_i\} \longrightarrow$ controllability index

Step 2:

$$\sigma_k := \sum_{i=1}^{k} d_i \qquad \sigma_1 = d_1,\quad \sigma_2 = d_1 + d_2,\quad \vdots,\quad \sigma_m = d_1 + d_2 + \cdots + d_m = n$$

e.g. $\begin{pmatrix} d_1 = 2 \\ d_2 = 2 \end{pmatrix}$ $\sigma_1 = 2$, $\sigma_2 = 4$

Step 3: Take L^{-1}

Set $q_k = \sigma_k^{th}$ row of L^{-1} for $k = 1, 2, \ldots m$

$k=1$: $q_1 = 2^{nd}$ row L^{-1}

$q_2 = 4^{th}$ row L^{-1}

$$Q = \begin{bmatrix} q_1 \\ q_1 A \\ \vdots \\ q_1 A^{d_1-1} \\ q_2 \\ q_2 A \\ \vdots \\ q_2 A^{d_2-1} \\ \vdots \\ q_m \\ \vdots \\ q_m A^{d_m-1} \end{bmatrix} \in \mathbb{R}^{n\times n}$$

Step 4: $\hat{A} = QAQ^{-1}$ $\hat{B} = QB$

Feedback design:

$$\begin{bmatrix} \tilde{A}_{11} & \tilde{A}_{12} & \tilde{A}_{13} \\ \tilde{A}_{21} & \tilde{A}_{22} & \tilde{A}_{23} \\ \tilde{A}_{31} & \tilde{A}_{32} & \tilde{A}_{33} \end{bmatrix} + \begin{bmatrix} \tilde{B}_1 \\ \tilde{B}_2 \\ \tilde{B}_3 \end{bmatrix} \begin{bmatrix} \tilde{F}_1 & \tilde{F}_2 & \tilde{F}_3 \end{bmatrix}$$

MIMO example $\{1,1,1,1\}$

$$\begin{tikzcd} \{e_1, e_2, \cdots\} \quad \mathbb{R}^3 \arrow[r, "A"] \arrow[d, "Q"'] & \mathbb{R}^3 \arrow[d, "Q"] \\ \{\ \} \quad \mathbb{R}^3 \arrow[r, "\hat{A}"] & \mathbb{R}^3 \end{tikzcd}$$

e.g.

$$\left(\begin{bmatrix} \tilde{A}_{11} & \tilde{A}_{12} \\ 0 & \tilde{A}_{22} \end{bmatrix} + \begin{bmatrix} \tilde{B}_1 \\ \tilde{B}_2 \end{bmatrix} \begin{bmatrix} \tilde{F}_1 & \tilde{F}_2 \end{bmatrix} \right) = \begin{bmatrix} \tilde{A}_{11} + \tilde{B}_1\tilde{F}_1 & \tilde{A}_{12} + \tilde{B}_1\tilde{F}_2 \\ \tilde{B}_2\tilde{F}_1 & \tilde{A}_{22} + \tilde{B}_2\tilde{F}_2 \end{bmatrix}$$

$AQ^{-1} = Q^{-1}\hat{A}$

$\Rightarrow QAQ^{-1} = \hat{A}$

$\tilde{F}$ Γ

Observability:

$$\left.\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t)\\ y(t) &= Cx(t) + Du(t)\end{aligned}\right\} \qquad y(t) = Ce^{At}x_0 + C\int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau + Du(t).$$

A state x_0 is said to be unobservable if the $\underbrace{\text{zero-input}}_{u(t)=0}$ $\underbrace{\text{response}}_{\text{output}}$ of the system is zero for all $t \geq 0$.

$$\left.\begin{aligned}\dot{x}(t) &= Ax(t)\\ y(t) &= Cx(t)\end{aligned}\right\} \quad \underline{\underline{y(t)}} = Ce^{At}\underline{\underline{x_0}}$$

$$x_0 \xrightarrow{\Psi} Ce^{At}x_0$$

Given $y(t)$ can we uniquely determine x_0?

x_0 can be identified unambiguosly from $y(t)$ $\Leftrightarrow \ker \Psi = 0$

Theorem:

$$\ker \Psi = \ker C \cap \ker CA \cap \cdots \ker CA^{n-1}$$
$$(\mathcal{N}_{CA})$$
$$= \ker \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

$\operatorname{img} \mathcal{C}_{AB} = \operatorname{img}\left[B \; AB \cdots A^{n-1}B\right]$

Proof: $\ker \Psi \subseteq \ker C \cap \ker CA \cdots \cap \ker CA^{n-1}$ ✓

Let $\underline{x_0} \in \underline{\ker \Psi} \Rightarrow \Psi(x_0) = 0 \Rightarrow Ce^{At}x_0 = 0 \quad \forall\, t \geq 0$

$t=0, \quad Cx_0 = 0 \Rightarrow x_0 \in \ker C \qquad \frac{d}{dt}\left(Ce^{At}x_0\right) = CAe^{At}x_0 = 0 \quad \forall\, t \geq 0$

$t=0: \quad CAx_0 = 0 \Rightarrow x_0 \in \ker CA$

$\vdots \qquad\qquad \vdots$

$CA^{n-1}x_0 = 0 \Rightarrow x_0 \in \ker CA^{n-1}$

$\therefore \underline{x_0} \in \ker C \cap \ker CA \cap \cdots \ker CA^{n-1}$

$\ker C \cap \ker CA \cdots \cap \ker CA^{n-1} \subseteq \ker \Psi$ ✓

$x_0 \in \ker C \cap \ker CA \cdots \cap \ker CA^{n-1}$

$$\Psi(x_0) = Ce^{At}x_0 = C\left(\Phi_0(t) + \Phi_1(t)A + \cdots \Phi_{n-1}(t)A^{n-1}\right)x_0$$

$$= \Phi_0(t)Cx_0 + \Phi_1(t)\,CAx_0 + \cdots \Phi_{n-1}(t)\,CA^{n-1}x_0$$

$$= 0$$

$x_0 \in \ker \Psi$

$$\ker \Psi = \ker C \cap \ker CA \cap \cdots \ker CA^{n-1}$$

$$\mathcal{N}_{CA} = \ker \underbrace{\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}}$$

(C,A) observability $\Longleftrightarrow$ $\underbrace{(A^T, C^T) \text{ controllability.}}$

$$\text{rank}\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} = n \Leftrightarrow \text{rank}\underbrace{\begin{bmatrix} C^T & A^TC^T & \cdots & (A^T)^{n-1}C^T \end{bmatrix}}_{= \mathcal{R}} \Leftrightarrow (A^T, C^T) \text{ controllable.}$$

The following statements are equivalent

(a) The controllability matrix $\mathcal{C}_{A^TC^T}$ has full row-rank, i.e. $\text{rank}(\mathcal{C}_{A^TC^T}) = n$

(b) (A^T, C^T) is controllable

(c) The matrix $[A^T - \lambda I \;\; C^T]$ has full row-rank $\forall \lambda \in \mathbb{C}$ (PBH-test)

(d) The matrix $W_t := \int_0^t e^{A^T\tau}C^TCe^{A\tau}d\tau$ is +ve def^n for any $t \in \mathbb{R}$.

(e) Let $\lambda \in \sigma(A^T)$ and x be it's corresponding left eigenvector. (Then $x^*A^T = x^*\lambda$)
Then $x^*C^T \neq 0$.

✓ (f) The eigen values of $(A^T + C^TF)$ can be freely assigned by suitable F (with the restriction that complex eigenvalues of $A^T + C^TF$ appears as conjugate pair.)